

RSDNet: Residual Shallow–Deep Attention Network for Single Image Dehazing

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Abstract—Single-image dehazing still faces challenges of uneven haze distribution and loss of information in reconstructed images. We develop RSDNet, a Residual Shallow–Deep Attention Network that restores high-quality haze-free images while maintaining low computational cost. The network integrates three complementary modules; Multi-Activated Attention Block (MAAB), Cascaded Residual Block (CRB), and Residual Attention Block (RAB) to capture shallow, hierarchical, and deep contextual information, respectively. MAAB promotes feature diversity through multiple nonlinear activations; CRB expands the receptive field and aggregates multi-scale information via cascaded residual connections, and RAB enhances deep contextual understanding with attention-based refinement. Trained with a hybrid Charbonnier+SSIM loss function, RSDNet achieves 2–3 dB PSNR gain and higher SSIM compared to state-of-the-art methods on both real-world (O-Haze) and synthetic (FRIDA) datasets. With only 5.96 M parameters and an average inference time of 63 ms (≈ 15.9 FPS), RSDNet delivers real-time dehazing while preserving natural colors and fine textures, achieving an effective balance between accuracy, efficiency, and perceptual quality.

Keywords—single-image dehazing, lightweight CNN, attention mechanism, residual learning

I. INTRODUCTION

Haze is a common atmospheric phenomenon that causes visibility impairment and compromises public safety [1]. Haze occurs due to the presence of fine particles in the air. These particles can be water droplets, sulfates, organic carbon compounds, dust, and soil particles [2]. These particles absorb and scatter light, causing images captured in hazy environments to lose vital details and appear blurred. This makes the dehazing a challenging task, and researchers are still trying to achieve satisfactory modeling systems in real-world scenarios [3][4]. Image dehazing has gained lots of interest recently in both academia and industrial communities [5]. The atmospheric scattering model (ASM) is widely used to describe this degradation process by relating the observed hazy image to the true scene radiance and transmission. It can be expressed as:

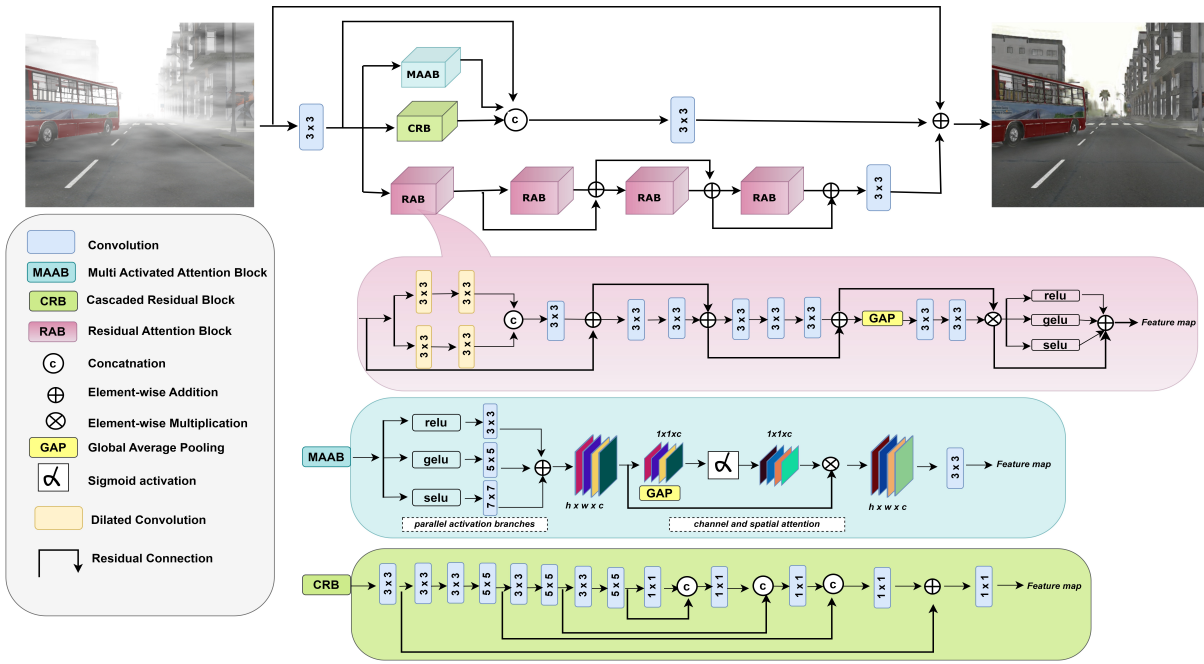
$$I = J(x)t(x) + A(1 - t(x)) \quad (1)$$

Here, $J(x)$ denotes the latent haze-free image, $t(x)$ represents the medium transmission map, and A corresponds to the global atmospheric illumination. Since both A and t are unknown in (1), the dehazing task becomes an ill-posed inverse



Fig. 1. Qualitative results of the proposed RSDNet method on the OH-Haze dataset, demonstrating its effectiveness in removing haze while preserving image details.

-rse problem [6]. Early image dehazing methods are based on prior based knowledge to estimate these parameters [7]. While these methods work well under prior assumptions, the recovered image could be distorted if the prior assumptions are not met. With the advancement of deep learning, learning-based image dehazing has achieved significant performance [8][9]. Although existing learning-based methods have achieved remarkable success, their performance is still limited by the loss of image detailed information such as color, hue and contrast information [10]. Also haze is not evenly distributed in all regions of image, so some pixels contain thin haze while others have thick haze. This makes the restoration process challenging. Recently, NLP studies motivated the vision researchers to include attention mechanism as a solid alternative to the convolutional designs and inherent global interaction between all pixels. Even though the attention mechanism is super effective in many image restoration tasks, including image dehazing [11], we still need to bypass it for hardware-friendly implementation. In image dehazing problem, attention mechanism can be employed to emphasize more on the regions which are affected by dense haze than those degraded by the light haze. Designing effective attention mechanisms can be challenging and might require extensive tuning. In order to realize the above an end-to-end single image dehazing network with multi-stage feature extraction modules is proposed. The dehazing method can directly restore the input hazy image to a haze-free image, as shown in Fig. 1, without estimating the parameters of ASM. In this paper, we introduce RSDNet, an innovative network designed for efficient and effective single-image dehazing. Our approach integrates multiple feature extraction modules to handle different levels of image details and haze density. The contributions of this paper are summarized as follows:
I) We introduce RSDNet, a lightweight and efficient end-to-end dehazing network designed to handle the complexity of haze removal while maintaining a low computational cost.
II) The proposed RSDNet comprises three modular compon



convolution to produce an intermediate fused representation. The output of RAB is then added to this fused feature map and combined with the input image, through a residual connection to generate the final dehazed image. The next subsections describe how each module functions, followed by an explanation of how all components work together in the complete RSDNet pipeline for a clearer understanding of the overall process.

3.1. Multi-Activated Attention Block (MAAB)

The MAAB is the first stage of RSDNet and works as a shallow feature extractor. It enhances the raw convolution output into a more meaningful, attention-enriched feature map. MAAB's goal is to turn simple linear features into richer, non-linear representations that better capture how haze varies across an image. Unlike conventional CNNs that rely on a single activation function, MAAB uses three activations ReLU, GeLU, and SeLU in parallel to increase diversity in feature learning. ReLU highlights strong edges and textures, GeLU handles smooth intensity transitions, and SeLU keeps learning stable through self-normalization. After fusing these activations, MAAB applies a dual attention mechanism: channel attention emphasizes the most important feature maps, while spatial attention focuses on the hazy regions of the image. Together, they help the network prioritize meaningful features and regions, resulting in clearer, more balanced, and visually natural dehazed images.

3.2. Cascaded Residual Block (CRB)

This module is the direct adaptation of the UNET architecture. Various computer vision tasks like segmentation, image reconstruction, generative modelling [20] considers Unet as their de facto network. There are several reasons for the popularity of this architecture:

Feature suction through skip connection helps in bringing old representation with current one. As a result, feature representation from the next layer is having more diverse and refined distribution.

Scaling at different resolution offer high or low frequency components to take part and contribute equally through it in encoder-decoder design.

Flexibility within inclusion of various deep learning design choices like residual, attention on pyramidal block, either in encoder, decoder or all layers. As a result, Unet-family expanded by adopting many effective feature extraction blocks.

Many image reconstruction studies partially or fully adopts Unet-family [11]. With the above benefits Unet also have specific dimension issues due to graph tracing limitations within regular DL libraries. For that reason, we can't easily incorporate Unet for all size of images as Unet-family explicitly needs up and down scaling operation. Our CRB module adopts Unet design without any scaling operation therefore, we don't face this issue of dimensions. As a result, we do loose the low and high frequency feature extraction but we partly address this by including different size of kernels (3x3, 5x5, 7x7). Due to that larger receptive field is present and we concatenated features from larger receptive field layers with smaller one. To avoid too large kernel, we include dilated operation and maintained above merging them. Hence, our CRB module retain efficient

feature extraction capability. After performing consecutive residual skipping concatenation with convolution operation, we exit CRB block with a simple 3 x 3 convolution operation.

3.3. Residual Attention Block (RAB)

The final module of RSDNet is RAB, which focuses on deep feature refinement and contextual understanding of an input image. The module will be learned to understand the interaction of complex features and the effect of long-range dependencies on haze formation of an image. This architecture contains four RAB modules which are organized in cascade form where the flow of information is gradual through a number of stages. In these modules, an output of a module is then connected to the input of the next module through a residual skip connection, which enables the flow of gradients smoothly and also the significant details of the image to be maintained. The cascading technique allows further refinements to be made at each of the stages where shallow features and deep semantics (complex and abstract) can be made clear and haze-free image. Every RAB has a merge-and-run unit, which divides the input features into two streams. One of them is applied with dilated convolutions to see the big picture, and the other of them employs regular convolutions to find fine detail. With this blend, the network is able to identify finer texture patterns as well as wider context information. These layers keep the training stable and keep the spatial information. Then, a lightweight channel attention mechanism is used to strengthen the most relevant features. After that, multiple non-linear activations (ReLU, GeLU, and SeLU) are used to enhance non-linearity. This integration of residual learning, attention, and multi-scale processing significantly contributes to the better performance of RSDNet in restoring clear and visually pleasing images.

3.4. Loss Function

To tune RSDNet during training, we use the combination of SSIM and Charbonnier as the loss function for the whole training process. Charbonnier Loss function is commonly used in computer vision tasks, especially for image reconstruction or regression problems [21]. It is a smooth approximation of the l_1 loss and is less sensitive to outliers while maintaining robust pixel-wise accuracy. Let's assume that the pixel intensity of the predicted dehazed image is I_{pred} , and the corresponding ground truth is I_{gt} , then the Charbonnier loss equation for dehazing is:

$$L_{CARE} = \sqrt{(I_{pred} - I_{gt})^2 + \epsilon^2}, \quad (7)$$

where ϵ is a small constant to prevent division by zero.

The SSIM loss improves structural similarity and preserves perceptual quality of reconstructed image. It calculated as:

$$L_{SSIM}(x,y) = 1 - \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}, \quad (8)$$

where, μ_x , σ_x^2 shows variance of input x , σ_{xy} is the covariance between images x and y , and c_1, c_2 are constants used to maintain stability.

The total loss is computed as:

$$L_{hybrid} = \alpha L_{CARE} + \beta L_{SSIM}(x,y), \quad (9)$$

where α and β are weighting coefficients that controls the relative contribution of each loss function.

3.5. RSDNet

Given an input hazy image $I_{\text{hazy}} \in \mathbb{R}^{h \times w \times 3}$, RSDNet performs a sequence of convolutional and attention-driven operations to reconstruct a clear output image I_{out} . The overall data flow of the network can be formulated as follows:

1) Initial features are extracted by passing I_{hazy} through a convolutional layer of 3×3 kernel followed by ReLU activation,

$$f_0 = \text{conv}(I_{\text{hazy}}) \quad (2)$$

where f_0 represents the shallow feature map obtained from the input.

2) The extracted features are fed simultaneously to three specialized modules as:

$$f_{\text{MAAB}} = A(f_0), f_{\text{CRB}} = C(f_0) \text{ and } f_{\text{RAB}} = R(f_0).$$

where A, C, and R denote the operations of the MAAB, CRB, and RAB modules, respectively.

3) The outputs from MAAB, CRB, and the initial convolution layer are concatenated along the channel dimension and refined by a 3×3 convolution:

$$f_{\text{fuse}} = \text{conv}([A \parallel C \parallel f_0]), \quad (3)$$

where $[\parallel]$ indicates channel-wise concatenation.

4) The output of RAB is added to the fused features and combined with the input image via a residual connection to ensure color and detail preservation:

$$f_{\text{residual}} = R + f_{\text{fuse}}, \quad (4)$$

$$I_{\text{out}} = f_{\text{residual}} + I_{\text{hazy}} \quad (5)$$

Thus, the complete RSDNet mapping can be summed up as:

$$I_{\text{out}} = I_{\text{hazy}} + f_{\text{RAB}} + \text{conv}([f_{\text{MAAB}} \parallel f_{\text{CRB}} \parallel f_0]) \quad (6)$$

This designs enables RSDNet to effectively learn both deep and shallow features by utilizing nonlinearity, residual and attention-based representations for recovering high quality haze free image with minimal computational cost.

IV. EXPERIMENTAL SETUP

These experiments were designed to assess both the quantitative and qualitative performance of the model against several state-of-the-art image dehazing methods.

4.1. Datasets

To evaluate the effectiveness of the proposed RSDNet, we carried out a series of experiments on two publicly available datasets: O-Haze [22], which contains real-world images, and FRIDA [23], a synthetic dataset.

O-Haze: The O-Haze dataset includes 40 outdoor scenes captured under real hazy conditions, each paired with a corresponding haze-free ground truth image. Every image has a resolution of 2833×4657 pixels. Following standard practice, the first 35 images were used for training and the remaining 5 for testing. The dataset offers a variety of real environmental settings, making it suitable for testing how well the model generalizes to natural haze conditions.

FRIDA: The FRIDA (Foggy Road Image Database) and FRIDA2 dataset consists of 84 synthetic hazy images depicting roadside scenes with different fog densities. Each image has a resolution of 640×480 pixels and is accompanied by its corresponding clear image. We used 80% of the dataset for training and 20% for testing. Because

of its synthetic nature and varying haze intensity, FRIDA provides a challenging benchmark for validating generalization under diverse conditions.

4.2. Evaluation protocol

The quantitative metrics include the Structural Similarity Index (SSIM), which is calculated between an image result and the ground truth, and the Peak Signal-to-Noise Ratio (PSNR), measured in decibels (dB) [1]. The restoration's integrity to the original image is measured by the score, which increases with higher scores.

4.3. Training settings

To increase data diversity, standard augmentation techniques such as random rotation and flipping horizontally and vertically are applied. All augmented images were then divided into non-overlapping patches of 64×64 pixels with a stride of 20 to ensure efficient training and better feature coverage. The model was trained using the Adam optimizer with default parameters. The learning rate was initially set to 1×10^{-4} and gradually decayed with the number of epochs. The batch size was fixed at 16, and gradient clipping with a norm of 1.1 was applied to maintain training stability. RSDNet was implemented in the TensorFlow Keras framework and trained on an NVIDIA Tesla V100 GPU equipped with Tensor Cores.

V. RESULTS AND DISCUSSION

In this section, we compare the proposed RSDNet with existing state-of-the-art (SOTA) image dehazing algorithms, both quantitatively and qualitatively, as summarized in Table 1 and Table 2. The quantitative evaluation on the FRIDA dataset indicates that RSDNet achieves consistently higher scores in both PSNR and SSIM, demonstrating its superior dehazing capability compared to previous approaches. We also evaluated our method against six SOTA dehazing algorithms, DCP, AOD-Net, DehazeNet, FastNet, FFA-Net, and EDN-GTM using the O-Haze dataset (Table 2). Although FastNet, FFA-Net, and EDN-GTM exhibit competitive performance, RSDNet achieves a better balance between accuracy, model complexity, and inference speed. In particular, EDN-GTM relies on a costly adversarial training phase to produce high-quality dehazed outputs. While adversarial learning can improve restoration performance, it significantly increases computation cost and often accumulates redundant gradients during multi-domain optimization. In contrast, RSDNet achieves comparable or better results with a simpler and more efficient architecture.

5.1. Qualitative and Efficiency Analysis

Visual comparisons (Fig. 3-4) further confirm that RSDNet effectively removes haze while maintaining natural color tones and texture fidelity. Competing methods often leave faint haze traces or introduce color distortions, whereas RSDNet outputs exhibit clearer object boundaries, balanced illumination, and perceptually realistic contrast. To assess computational performance, the average inference time of RSDNet was measured across multiple test images. The model processes each image in 63 milliseconds, corresponding to a throughput of approximately 15.9 frames per second (FPS). This confirms that RSDNet operates within the real-time processing range while preserving high perceptual and structural quality. With only 5.96 million

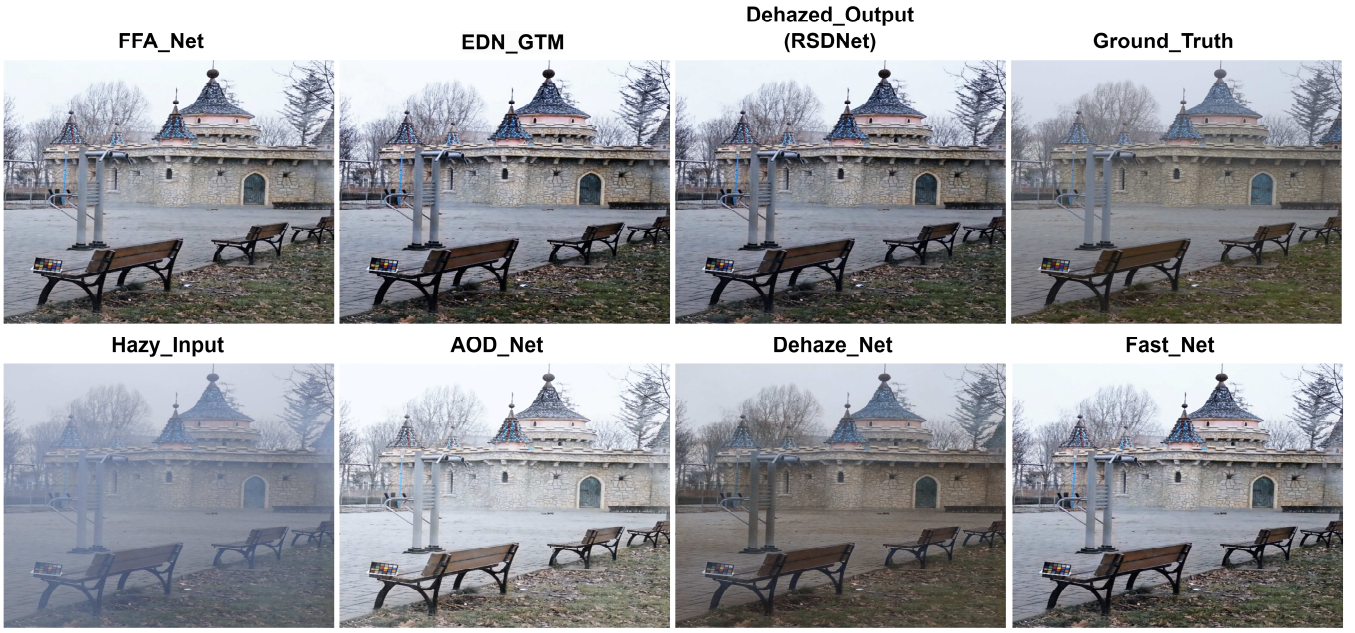


Fig. 3. Qualitative comparison on O-HAZE dataset. The images demonstrate the performance of RSDNet (right) compared to other methods: AODNet, DehazeNet, and FFA-Net (left to right). Notice the improved color retention and clarity in the RSDNet results.

parameters, RSDNet remains computationally efficient compared to heavier architectures such as FFA-Net (68.33M parameters). Overall, RSDNet provides a favorable trade-off between accuracy, speed, and model size, enabling robust dehazing without any pre- or post-processing steps. The network generalizes well across various haze densities, producing consistent and perceptually convincing results suitable for real-time and embedded applications.

5.2. Ablation Study

We performed an ablation study by removing its three core components (MAAB, CRB, RAB) one-by-one, in order to validate the contribution of each module in the design of RSDNet (see Table 3). Each variant was trained under identical settings by using OH dataset. FLOPs are estimated proportionally to parameter count (baseline: 5.96 M $\rightarrow$ 15.9 G FLOPs).

Effect of MAAB: The goal of MAAB is to enrich feature representation through multiple activation functions with adaptive weighting. This approach enables the network to capture diverse non-linear relationships within haze regions, extract sharper edges and enhance local contrast. The removal of MAAB results in noticeable degradation in performance, especially in PSNR, confirming its role in fine-detail recovery.

Effect of CRB: The CRB allows the network to effectively reconstruct large-scale haze patterns while preserving gradient information. Excluding the CRB leads to visible loss of structure and reduced SSIM, describing its role in maintaining spatial consistency.

Effect of RAB: The purpose of RAB is to improve global color balance and overall perceptual quality of dehazed images. It achieves through dual-path attention (spatial and channel) mechanisms, by focusing on informative regions and suppressing redundant information in feature maps. Notably, excluding the RAB causes the largest performance drop in terms of PSNR and SSIM, revealing its crucial role in RSDNet working.

When all three modules are combined but only a single loss function (SSIM) is used, the visual performance drops, showing lower SSIM and PSNR scores. Overall, the ablation results show that integrating all three modules significantly improves the quantitative results, while adding SSIM loss alongside Charbonnier loss further enhances the visual quality without affecting inference speed.

TABLE 1. Quantitative comparison of the proposed model with existing methods on the FRIDA dataset.

Model	PSNR	SSIM
Yuan et al. [24]	12.76	0.432
Tai et al. [25]	13.94	0.791
Tarel et al. [26]	16.30	0.231
Zahra et al.[27]	18.23	0.823
Proposed	22.86	0.935

TABLE 2. Quantitative results of our proposed model on the OH-Haze dataset.

Method	PSNR	SSIM	Param(M)	FLOPs	Inference (ms)
RSDNet w/o CRB	20.53	0.78	5.85	15.61	52
RSDNet w/o MAAB	20.10	0.75	4.82	12.86	49
RSDNet w/o RAB	17.00	0.69	0.44	1.17	8
RSDNet(CRB,MAAB,RAB)	23.12	0.82	5.96	15.90	63
w/o SSIM	25.20	0.84	5.96	15.9	63

TABLE 3. Impact of CRB, MAAB and RAB modules in RSDNet on the OH-Haze dataset.

Method	PSNR	SSIM	Param(M)	FLOPs	Inference (ms)
RSDNet w/o CRB	20.53	0.78	5.85	15.61	52
RSDNet w/o MAAB	20.10	0.75	4.82	12.86	49
RSDNet w/o RAB	17.00	0.69	0.44	1.17	8
RSDNet(CRB,MAAB,RAB)	23.12	0.82	5.96	15.90	63
w/o SSIM	25.20	0.84	5.96	15.9	63

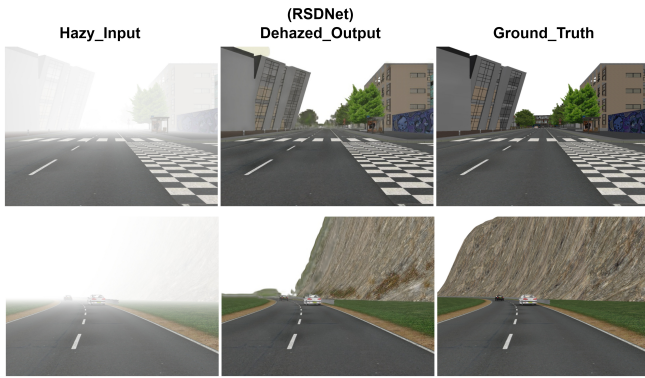


Fig. 4. Visual results of the RSDNet on the FRIDA dataset.

VI. CONCLUSION

In this work, we introduced RSDNet, an end-to-end multi-stage feature extraction network designed for single-image dehazing. The model was built to overcome three major challenges: uneven haze distribution, loss of structural information, and color distortion in restored images. To address these, the MAAB module enhances feature diversity through multiple activation functions, helping the network adapt to spatially varying haze. The CRB module gathers information from different scales to recover fine textures and global structures, while the RAB module uses attention mechanisms to refine color balance and brightness across the image. Together, these modules enable RSDNet to generate clear, visually natural results without increasing computational complexity. Evaluations on both synthetic and real-world datasets show that RSDNet performs on par with or better than heavier state-of-the-art models. In the future, we plan to expand this framework toward a unified restoration model capable of handling multiple degradations such as rain, dust, and fog.

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